GE Healthcare

3.0T Head, Neck, Spine Coil For GE 3.0T MRI Systems

Service Manual

DOC0639637
Rev 4
P/N 5367722

Revision History

Date	Rev	Change Description
01DEC2009	1	Initial Release
09DEC2009	2	Update Illustration 13 in section 5.5.10
07JAN2010	3	Updated step 1 in section 2.1. Corrected HNU part number in section 1.1
11FEB2010	4	Added Unified phantom instructions and differentiated Unified phantoms with Legacy phantoms to Section 3. Updated format of service language disclaimer. Clarified figure numbers throughout. Clarified callouts and arrows for better black and white printing throughout. Pg 22 to 27-clarified the set-up steps of the MCQA test using Legacy phantoms.

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Rev. 6 November 12, 2009

Language Policy Service Documentation (DOC0620154 Rev 1)

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DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

Table of Contents

ATTENTION!

Installation of this coil MUST be performed by a GE field engineer. Please contact your field engineer to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

1 3T HNS Coil Introduction

1.1 Product Identification

To identify the 3T HNS Coil, refer to the part numbers and descriptions that are provided on the coil rating plates (refer to Table 1).

Table 1: Coil Identification

GEHC part number	5338925	5336485	5338938
Description	“3T HNS HNU w/o cable”	“3T HNS TLU w/o cable”	“3T HNS NCU”

NOTE: Each of the coil’s (3) subcomponents contain its own label with part numbers and descriptions as shown in Table 1. The HNU FRU consists of the Posterior, Anterior, and Horseshoe Adapter.

Figure 1: The 3T HNS Coil (HNU with Posterior and Anterior, NCU, and TLU)

1.2 Compatibility

The 3T HNS Coil is available in three configurations: MR750, HDxt Forward Production (FP) and HDxt Upgrade (Upg), which differ only in system connector and cabling. FP configuration is for LPCA's that have only upper A, B, and C ports on the LPCA. Upgrade configuration has upper A and B ports and a lower legacy port.

1.3 Shipping Contents

NOTE: All pieces should be visually inspected for damage (i.e. bent pins, holes, cracks) prior to use.

Refer to 3T HNS Operator Manual for shipping contents.

1.4 Related Documentation

3T HNS Operator Manual, GE Part Number 2423588.

1.5 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

2 3T HNS Coil Installation Procedure

2.1 3T HNS Coil Installation Procedure for HDx systems

1. Make sure that the latest software build is loaded on the system prior to installing the HNS coil. The minimum software build is 15.0_M3 or 15.0_M4. (**Note:** This procedure does not apply to 15.0_M4A)

2. Open the Config File Manager (CFM) tool from the [Utilities] tab of MR CSD.

3. Go to the Coil config section and:

   - For a Forward Production (FP) system with M3 software, remove the following 5 HNS configurations:
     <GE_HNSTLsNoUse>
     <GE_HNSCTL2>
     <GE_HNSBrPlCsII>
     <GE_HNSCTBrain2>
     <GE_HNS NVfullFP>
   - For a Forward Production (FP) system with M4 software, remove the following 5 HNS configurations:
     <GE_HNSTLsNoUse>
     <GE_HNSCTL2>
     <GE_HNSCTBrain2>
     <GE_HNSBrPlCsII>
     <GE_HNS NVfullFP>
   - For an Upgrade system, remove the following 6 HNS configurations:
     <GE_HNSTLsNoUse>
     <GE_HNSCTBrain2>
     <GE_HNSBPCsII4>
     <GE_HNSBPCsII3>
     <GE_HNSCsICTs0>
     <GE_HNSBPCsII1>

   Remove the HNS configurations by doing the following. See Figure 2.
   a. Select the coil from the coil name
   b. Then select **Remove Coil** in the lower right side of the GUI
   c. Select **Yes**
   d. Repeat for the 5 mentioned coils for FP system, or for the 6 mentioned coils for the Upgrade system
   e. Select **Save Changes**, then select **Yes**
   f. Select **Quit** to exit Config File Manager.

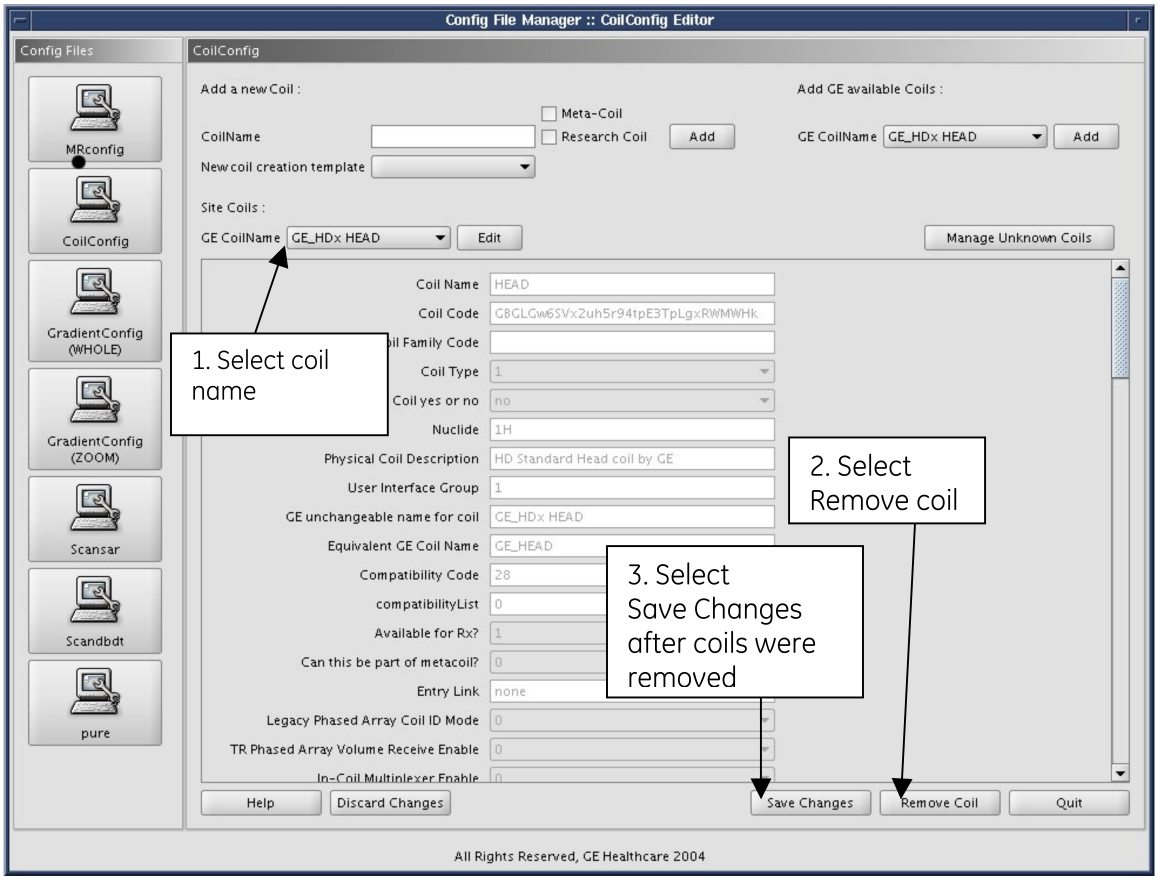


Figure 2: Removing coil config

4. Put the Coil CD (PN 5325975) into External / Internal DVD Drive.

5. Open a command window and type the following commands:
   su
   < root password >
   mount /mnt/cdrom
   /mnt/cdrom/coil_prescan_update

6. Launch CSD, go to [Utilities], Select [Install Coil CD].

7. Select [Click here to start this tool], an xterm window will open.

   If the CD has already been installed on the system, you will get a message that states the packages are already on the system disk. Eject CD and proceed to step 8.

   It may take up to 10 minutes for the window to open and for the CD to be ejected automatically based on system configuration. A message in the xterm window should appear stating "Coil Media Installation Successful". At this time the CD is ejected. Remove it then close the xterm window.

8. Reboot the System and login as SDC.

Note: Each physical configuration of the HNS coil must be installed separately for the system to recognize it. **Connecting all the coil components together will not result in the coil getting completely configured.** Setups 1 through 4 show the valid physical coil configurations for installation.

Note: Ensure that the coil components have a snug fit with each other.

If the coil could not be configured successfully, **DO NOT ATTEMPT TO INSTALL THE COIL MANUALLY WITH CONFIG FILE MANAGER OR VIA ANY OTHER METHOD**, please call the online center for additional information.

9. Connect the coil to the LPCA as shown in Setup 1. A pop-up will appear on the monitor after connecting the coil stating, " A new coil has been connected to the system, Do you want to configure?" Select **YES**. Then select **OK**.

10. After proper configuration of setup 1, remove the coil cable from LPCA, prior to changing coil configuration for setup 2.

11. Repeat the steps 9 and 10 for setups 2 through 4.

- **Setup 1:** Head Neck Unit only (Posterior + Anterior)

Note: The pictures show Head Neck Unit in FP configuration – single connector. Connect cable to B-port. The MR750 configuration will have a "P" cable. The Upgrade configuration will have an additional cable that is connected to the legacy port.

- **Setup 2:** Head Neck Unit only (Posterior + Horseshoe Adapter)

- **Setup 3:** Posterior + Anterior + TL + NCU

Note: Connect cable from head section to B-Port, and TL section to A-Port.

The FP configurations are shown in the pictures. The Upgrade configuration will have an additional cable that is connected to the Legacy port.

- **Setup 4:** Posterior + Horseshoe Adapter + TL + NCU

12. After all four coil configurations have been entered; reboot the system.

13. Prescribe new exam. Enter the patient information and click on the coil selection button. Verify that the HNS coil is present when clicking on the Currently Connected button under Coil Type on the scanner user interface. For example, if you use Setup 1 you will see the clinical coil modes as shown in
14. Figure **3**.

In forward production systems, the presence of coil entries like those shown is a confirmation that the coil installed properly. If you do not see the HNS coil modes, **DO NOT ATTEMPT TO INSTALL THE COIL MANUALLY**, **call the online center for additional information**.

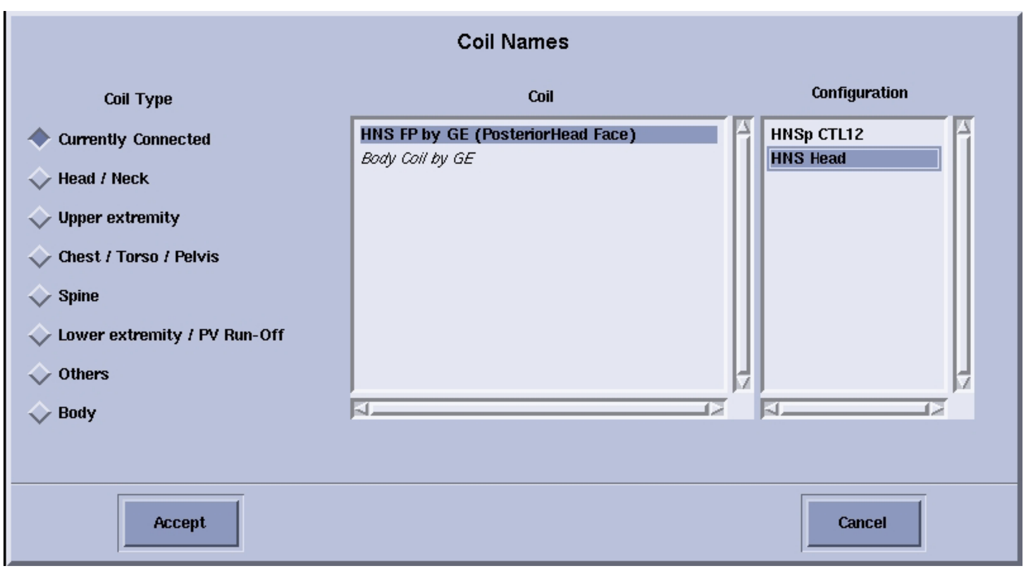


Figure 3: Confirmation of proper configuration

15. Perform MCQA test on coil.

Reminder: Complete Product Locator information for serialized components. Failure do this may result in failure of your site to receive future FMI's.

2.2 3T HNS Coil Installation Procedure for MR750 system

NOTE: Make sure that the latest software build is loaded on the system prior to installing the HNS coil. The minimum software build is 20.0_M4 or 20.1_IB1.

1. Put the Coil CD into External / Internal DVD Drive.
2. Open a command window and type the following commands:
   su
   < root password >
   mount /mnt/cdrom
   /mnt/cdrom/autorun
3. Enter "i" to install new coils from this disc.
4. After installation is complete, type "eject" to remove the Coil CD.
5. Close the command window.
6. Perform an MCQA Test on the coil.

3 3T HNS Coil MCQA Test

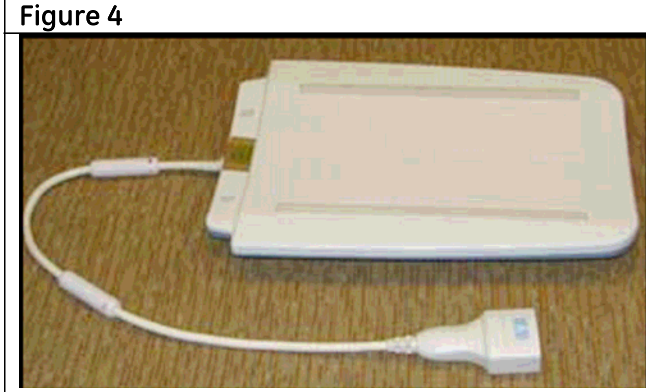
Follow this procedure to perform the MCQA test.

NOTE:
• The coils must be tested in 3 configurations:
1. **The TL unit only**
2. **Posterior + NCU + Horseshoe Adapter + TL**
3. **Posterior + Anterior**

This is done to ensure modularity in design and in diagnostics.
• The MCQA tool can ONLY run on the above mentioned individual coil configurations. The MCQA tool can NOT run on all coil configurations at the same time. MCQA will not run and will report that the configuration is invalid if an attempt is made to scan with an unsupported combination of coils.

NOTE:
1.5T phantoms contain green solution.
3.0T phantoms contain pink solution.

3.1 Record the coil serial numbers of the following coils:

 **Figure 4**	HNU Serial # ________________________ (The serial #s of Posterior, Anterior, and Horseshoe Adapter should all match)
 **Figure 5**	TL Serial # __________________________

3.2 Legacy Tools and Test Equipment

- 3T HNS Head Neck Unit Phantom kit (SiOil) – one cylinder, one brick, qty. 1 ea
  - GEHC Part Number: 2413435 & 2413436
- 3T HNS TL Unit Phantom (SiOil) – one large brick phantom, qty. 2
  - GEHC Part Number: 2381682-10
- 3T HNS HNU phantom positioner, 2420112, qty. 1

3.3 Required Conditions

The appropriate coil configuration files must be installed to run this tool.

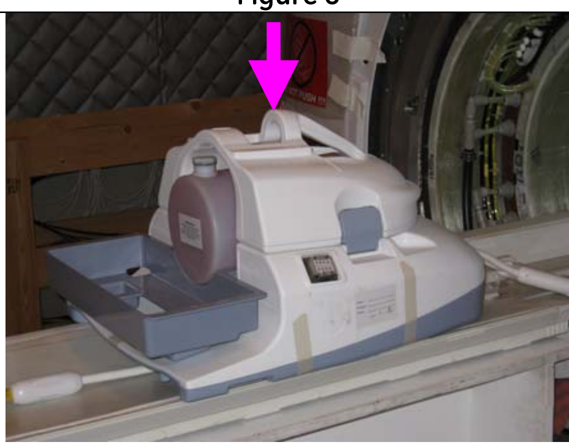
3.4 3.0T HNS Posterior Head Face using Legacy phantoms

3.4.1 Tools required

Description	Part Number	Qty
3T HNS Head Neck Unit Phantom kit (SiOil) – one cylinder	2413435	1
3T HNS HNU phantom positioner	2420112	1

3.4.2 Coil-Phantom Setup Procedure

Visual	Description
 **Figure 6**	**Posterior Head Face section** Place Posterior Head Face section on the table as shown in Figure 6.

 **Figure 7**	Place the 3.0T HNS phantom positioner in the coil. Place the Legacy cylinder phantom into the phantom positioner, and then carefully align the face component part of the coil with the connector pins over the Posterior head face section and latch both the coil sections into place as shown in Figure 7. (The scanner will not operate if the coil sections are not latched correctly).
 **Figure 8**	Connect the appropriate cables to the LPCA (Example of the Upgrade Configuration Posterior + NCU + Horseshoe Adapter + TL shown in Figure 8, other configurations may have different connections).
 **Figure 9**	Landmark the coil at the inferior edge of the arch as shown in Figure 9. Press advance to scan. Refer to Section 3.10 of this document to run the MCQA Tool with the Legacy Phantom Set.

3.5 3.0T HNS Posterior Head Chest Horseshoe TL using Legacy phantoms

3.5.1 Tools required

Description	Part Number	Qty
3T HNS Head Neck Unit Phantom kit (SiOil) – one cylinder	2413435	1
3T HNS Head Neck Unit Phantom kit (SiOil) – one brick	2413436	1
3T HNS TL Unit Phantom (SiOil) – one large brick phantom	2381682-10	2
3T HNS HNU phantom positioner	2420112	1

Coil-Phantom Setup Procedure

Figure 10

Place the Posterior Head Face section and TL section of the coil on the patient table. Place the phantom positioner in the Head section. Place and latch the Horseshoe on the Head section. Place the Legacy cylinder phantom into the phantom positioner, and place the brick Head Neck Unit phantom handle down into the phantom positioner. See Figure 10.

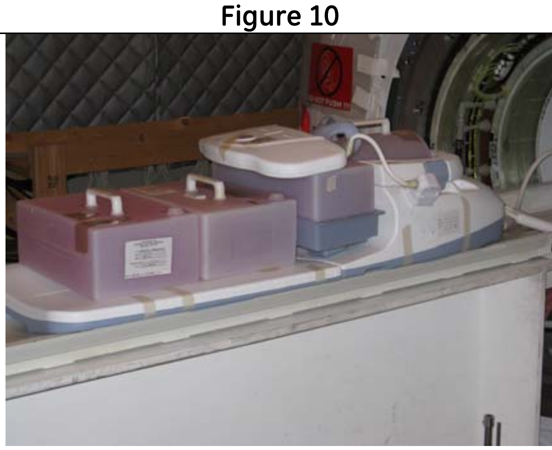
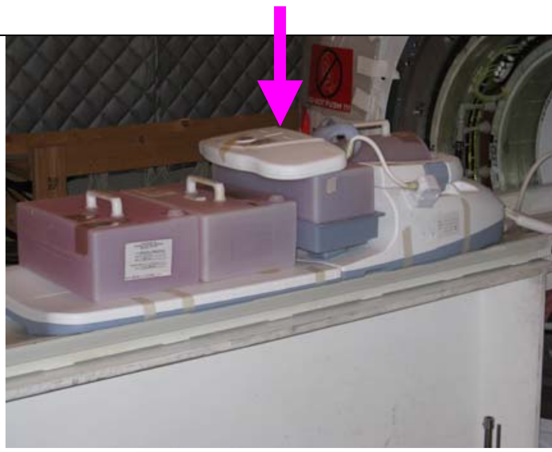
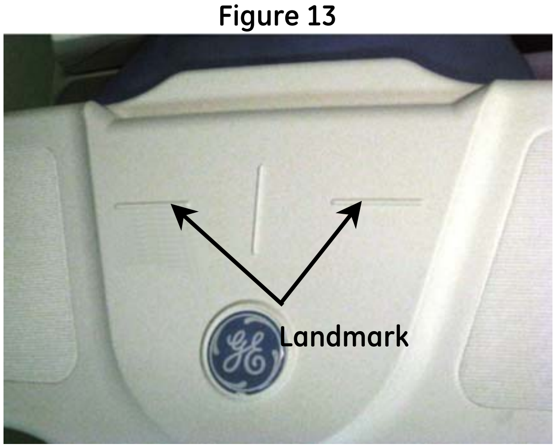


Figure 11

Place both TL Unit phantom bricks on the TL as shown in Figure 11.

Connect the chest part to the posterior Head section as shown in Figure 11, and complete the setup.

Ensure that the two TL bricks are in contact with each other and ensure the superior TL brick is in contact with the phantom positioner.

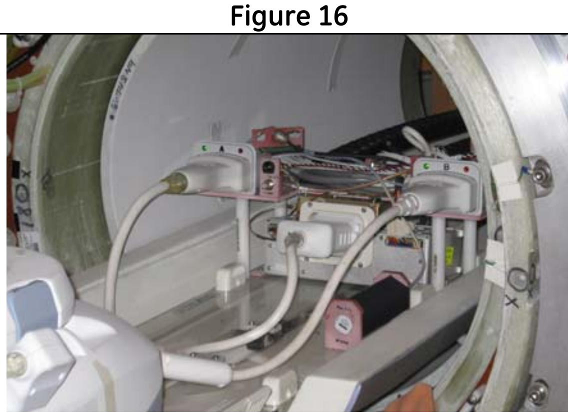
 **Figure 12**	Connect the appropriate cables to the LPCA (Example of the Upgrade Configuration Posterior + NCU + Horseshoe Adapter + TL shown in Figure 12, other configurations may have different connections).
 **Figure 13**  **Figure 14**	Landmark on the markings over the chest section of the coil as indicated in Figure 13 and Figure 14. Press advance to scan. Refer to section 3.10 of this document to run the MCQA Tool with the Legacy Phantom Set.

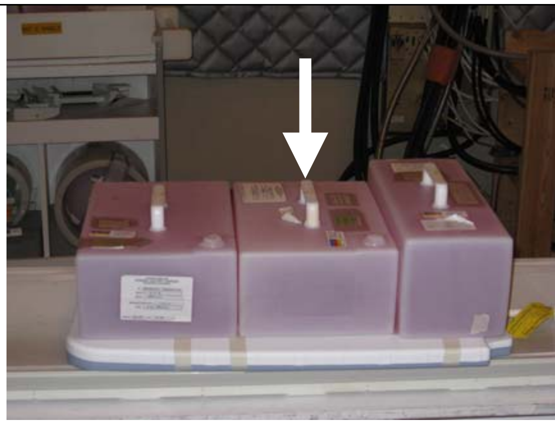
3.6 3.0T HNS TL using Legacy phantoms

3.6.1 Tools required

Description	Part Number	Qty
3T HNS Head Neck Unit Phantom kit (SiOil) – one brick	2413436	1
3T HNS TL Unit Phantom (SiOil) – one large brick phantom	2381682-10	1

3.6.2 Coil-Phantom Setup Procedure

Visual	Description
 **Figure 15**	Place the TL section of HNS coil on the table as shown in Figure 15.
 **Figure 16**	Place both brick TL phantoms and the brick Head, Neck Unit phantom on the TL section of the coil and align them properly in S/I direction as shown in Figure 16.
 **Figure 17**	Connect the appropriate cables to the LPCA (Example of the Upgrade Configuration Posterior + NCU + Horseshoe Adapter + TL shown in Figure 17, other configurations may have different connections).

 **Figure 18**	Landmark as shown in Figure 18. For the TL only setup, the handle of the middle phantom must align with the digit 5 on the side of the TL and the landmark line must pass through the handle/digit 5 as shown in Figure 18. Press advance to scan. Refer to section 3.10 of this document to run the MCQA Tool with the Legacy Phantom Set.

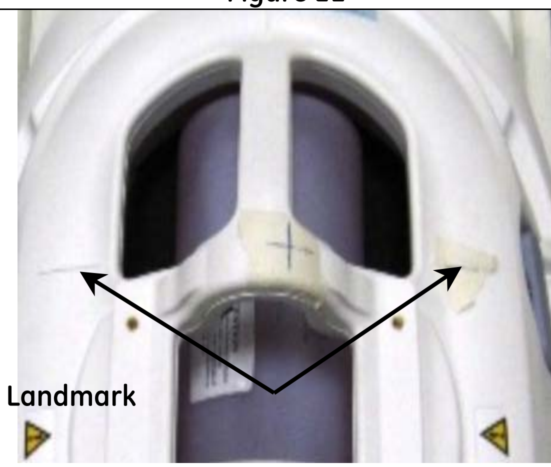
3.7 3.0T HNS Posterior Head Face using Unified phantoms

3.7.1 Tools required

Description	Part Number	Qty
Large Cylindrical Unified Phantom, SiOil	5342679-2	1
Phantom Unified positioner, HNS	5344844	1

3.7.2 Coil-Phantom Setup Procedure

Visual	Description
 **Figure 19**	**Posterior Head Face section** Place Posterior Head Face section on the table as shown in Figure 19.
 **Figure 20**	Place HNS positioner and then place large cylindrical unified phantom in the positioner as shown in Figure 20.

 Figure 21	Carefully align the face component part of the coil with the connector pins over the Posterior head face section and latch both the coil sections into place as shown in Figure 21. (The scanner will not operate if the coil sections are not latched correctly).
 Figure 22	Connect the coil to respective port of the system LPCA. Landmark the coil on the markings present on the face component indicated through arrows as shown in Figure 22 and press advance to scan. Refer to Section 3.10 of this document to run the MCQA Tool with the Unified Phantom Set

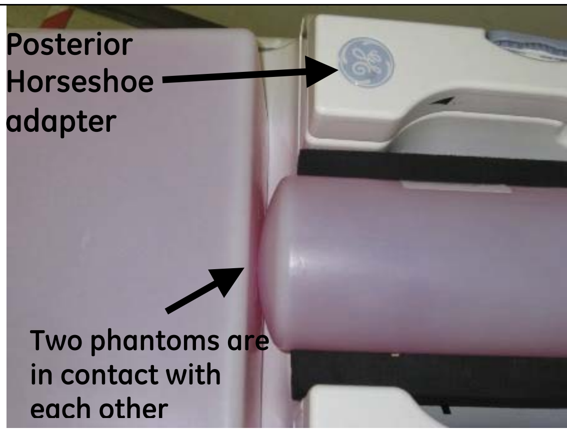
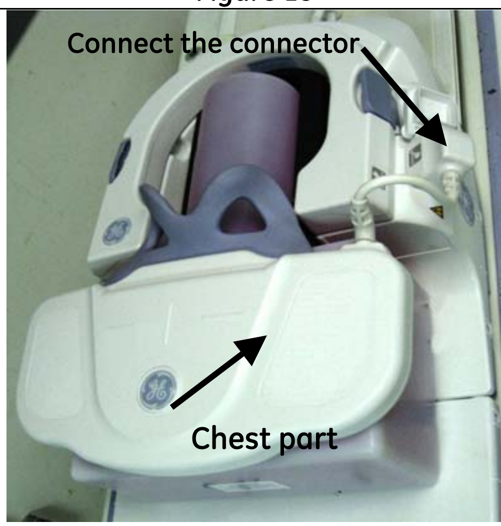
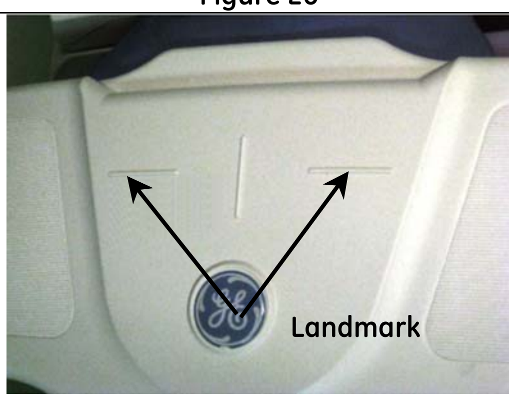
3.8 3.0T HNS Posterior Head Chest Horseshoe TL using Unified phantoms

3.8.1 Tools required

Description	Part Number	Qty
TL Unified Phantom, SiOil	5343347-2	1
Large Cylindrical Unified Phantom, SiOil	5342679-2	1
Phantom positioner, HNS	5344844	1

Coil-Phantom Setup Procedure

 **Figure 23**	**Posterior Head Chest Horseshoe TL section** Place the Posterior Head Face section and TL section of the coil on the patient table. (Figure 23) Place the phantom positioner as shown in the figure.
 **Figure 24**	Place the Large cylindrical unified phantom as shown in Figure 24.

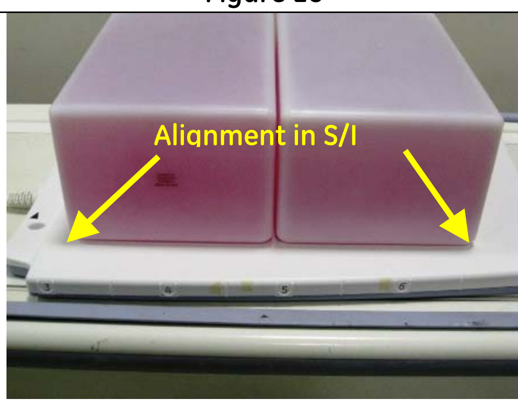
 **Figure 25**	Place the TL unified phantom over the junction of the TL section and head components and carefully latch the Horseshoe adapter of the face component into the connectors as shown in the Figure 25. Ensure that TL unified phantom is in contact with cylindrical unified phantom as shown in Figure 25.
 **Figure 26**	Connect the chest part to the posterior Head section as shown in Figure 26. The complete setup of this point is shown in Figure 26.
 **Figure 27**	Landmark on the markings over the chest section of the coil as indicated in Figure 27. Press advance to scan. Refer to section 3.10 of this document to run the MCQA Tool with the Unified Phantom Set.

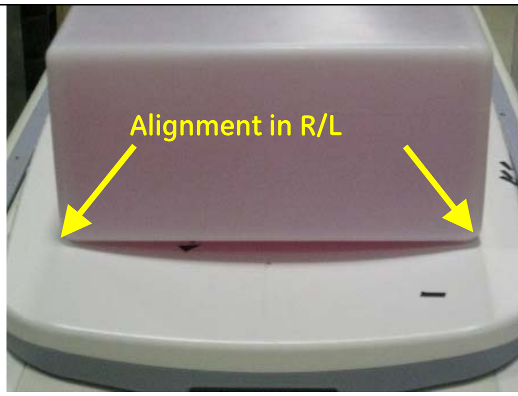
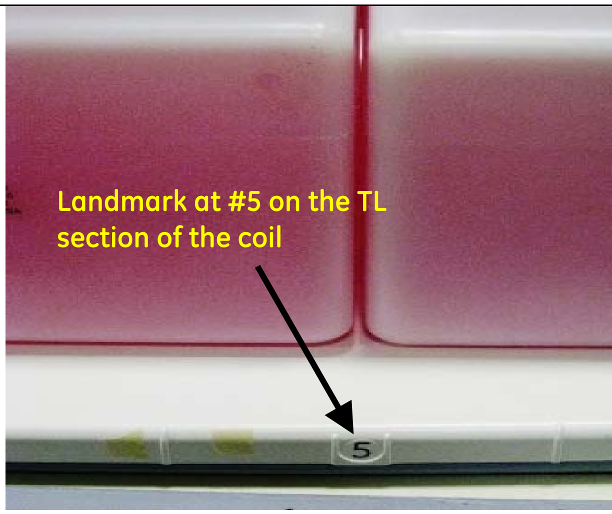
3.9 3.0T HNS GE TL using Unified phantoms

3.9.1 Tools required

Description	Part Number	Qty
TL Unified Phantom, SiOil	5343347-2	2

3.9.2 Coil-Phantom Setup Procedure

Visual	Description
 Figure 28	**TL section** Place the TL section of HNS coil on the table as shown in Figure 28.
 Figure 29	Place TL unified phantoms on the TL section of the coil and align them properly in S/I direction as shown in Figure 29.

 Figure 30	Align TL unified phantoms properly in R/L direction of the TL section of the coil as shown in Figure 30.
 Figure 31	Connect the coil to respective port of system LPCA. Landmark on the digit 5 present on the side of the TL section of the coil as shown in Figure 31 and press advance to scan. Refer to section 3.10 of this document to run the MCQA Tool with the Unified Phantom Set.

3.10 Steps to run MCQA

3.10.1 Warning: All RF coil related tests must be run on a system that is well-calibrated and passing all system tests (The system should have passed "Install In Spec" (IIS)), especially white pixel, correlated noise, and MCR (Multi-Coil-Receive) tool.

1. From Common Service Desktop (CSD) go to Service Browser and select **[Image Quality]**, "Multi-Coil QA Tool" and then "Click here to start this tool" as shown in Figure 32.

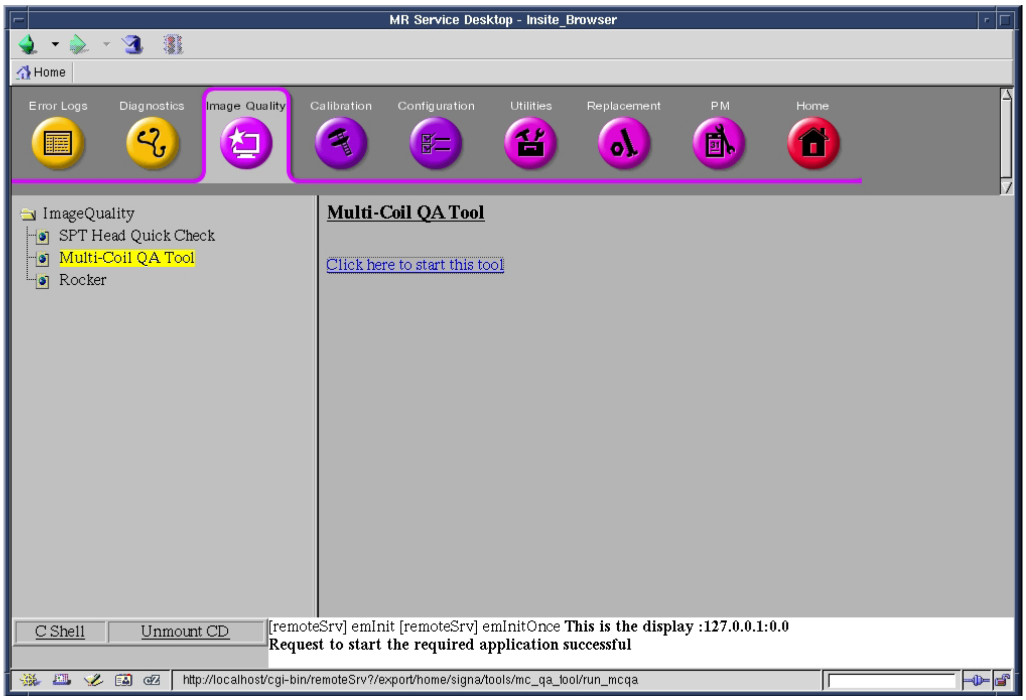


Figure 32: MCQA from the Service Desktop

2. Upon start-up, the coil that is connected to the system gets selected in the *Current coil* field as shown in Figure 33.

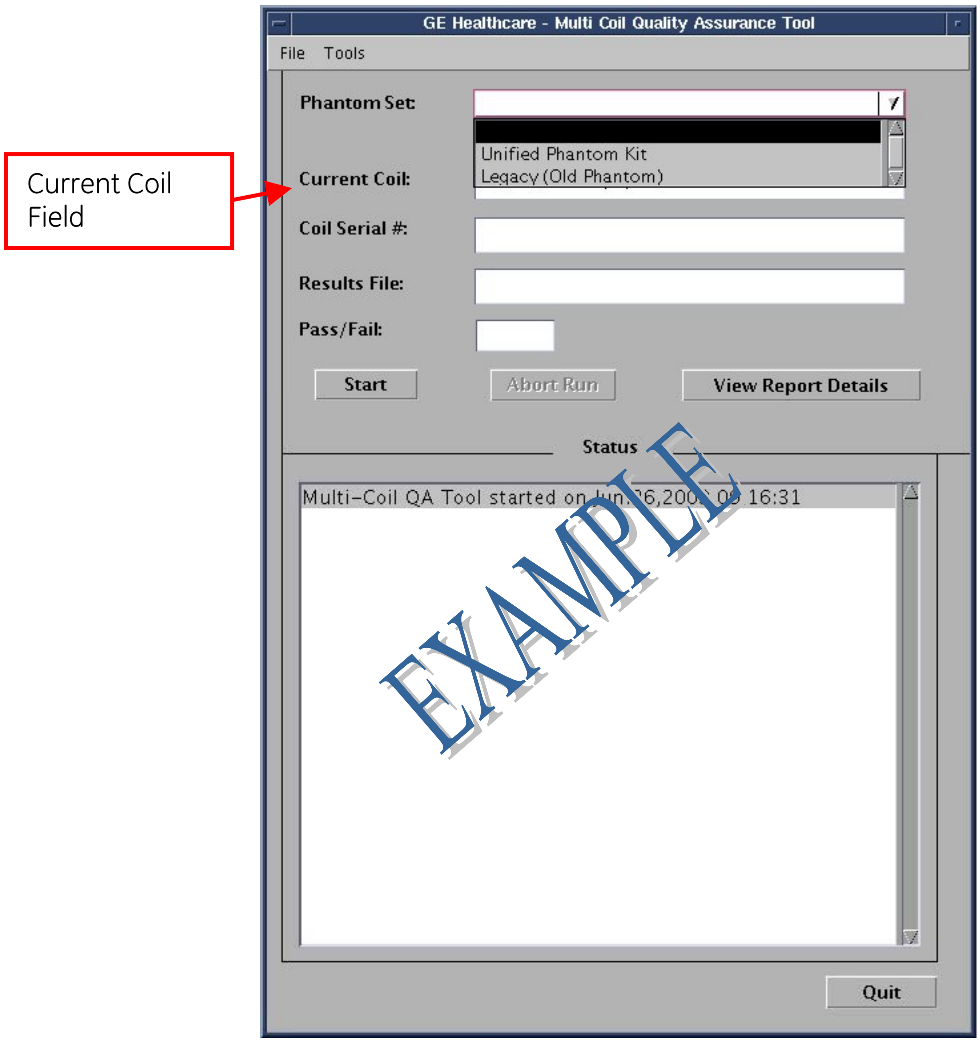


Figure 33: MCQA Tool Dialog box

3. A pop-up displays requesting you to, **Please select correct Phantom Set** as shown in Figure 34. Press OK.

Figure 34

4. If a Legacy Phantom Set is being used, then select “Legacy (Old Phantom)” from the dropdown menu that is next to the Phantom Set field. If “Unified Phantom Set” is being used then select “Unified Phantom Kit” from the drop down menu (Figure 35).

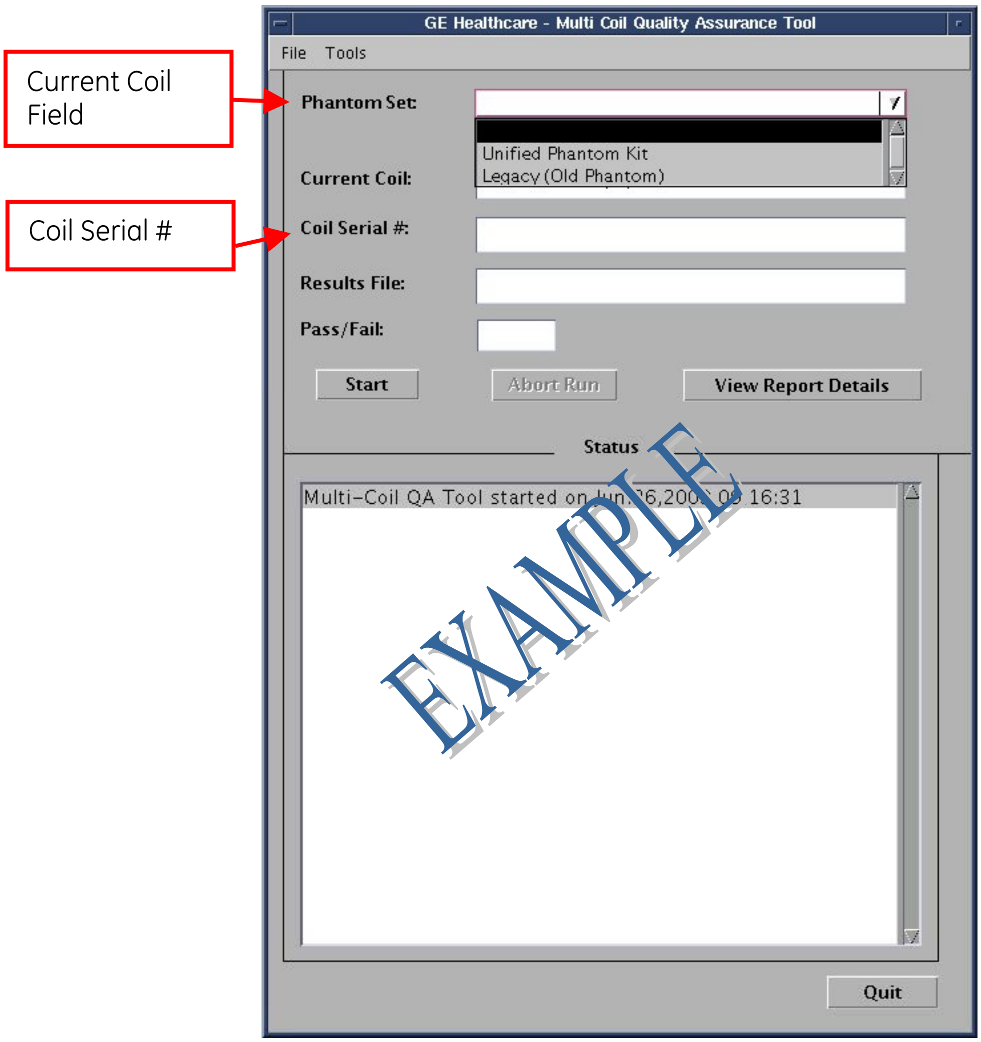


Figure 35

5. The current coil field will be automatically filled in, based on the CoilID of the coil connected to the LPCA. Enter the serial number of the coil being tested in the Coil Serial number field. See Figure 35.

6. Click on **[Start]** to begin the automated test as shown in Figure 36 Depending on the number of test locations (complexity of the coil) the test may take from 3 to 20 minutes.

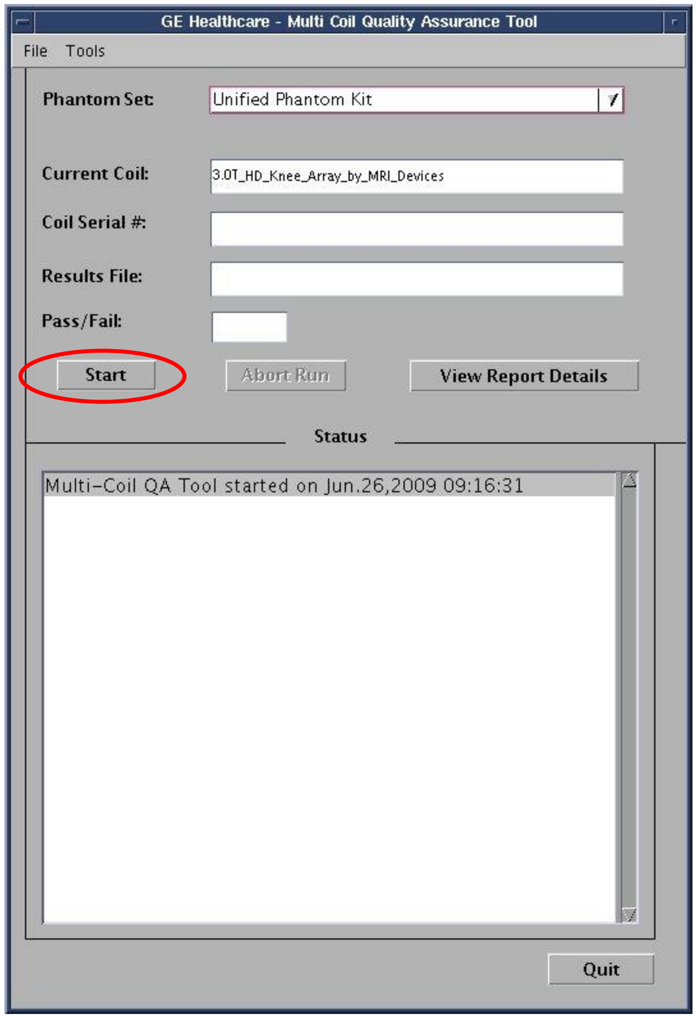


Figure 36

7. A pop-up displays asking if you want to continue. Click on **[Yes]**. See Figure 37.

NOTE: If a Legacy Phantom Set is being used, then a similar pop-up comes up for "Legacy (Old Phantom)."

Figure 37

8. Upon start-up, a Note stating, “Phantom placement and coil landmarking are critical for repeatable results” will appear. If the landmark has been set correctly and there are no air bubbles in the phantom, click **[Yes]** to continue. (Figure 38)

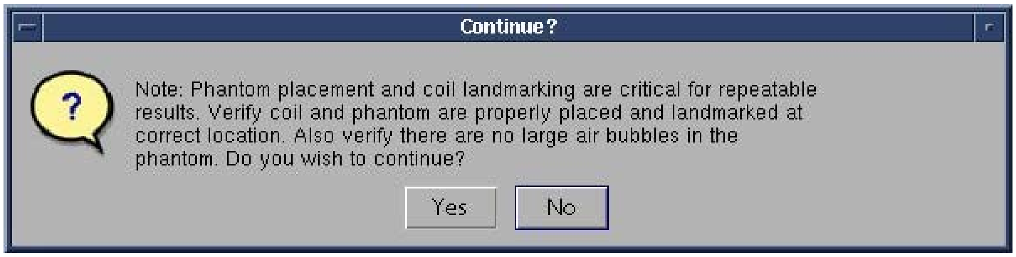


Figure 38

Note: The Status window of the MCQA Tool GUI will continuously update to give information on what the tool is doing at any point in time. A time bar (Figure 39) will appear, showing approximate total test time, elapsed time and percent complete. Once the test has completed, the name of the Results file will be listed, along with the Pass/Fail status and the results of the test in the MCQA Tool GUI.

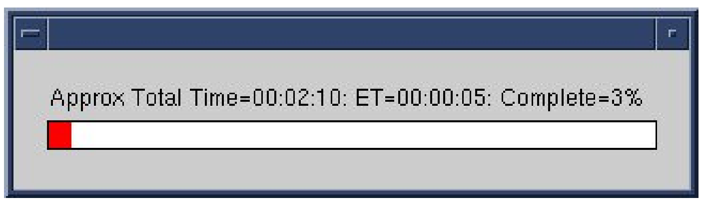


Figure 39

9. When the test is complete, test results display on the screen (Figure 40). The PASS/FAIL status shows PASS if all coil elements are functioning properly. If any coil elements display “FAIL”, call the GE Service Representative for coil repair.

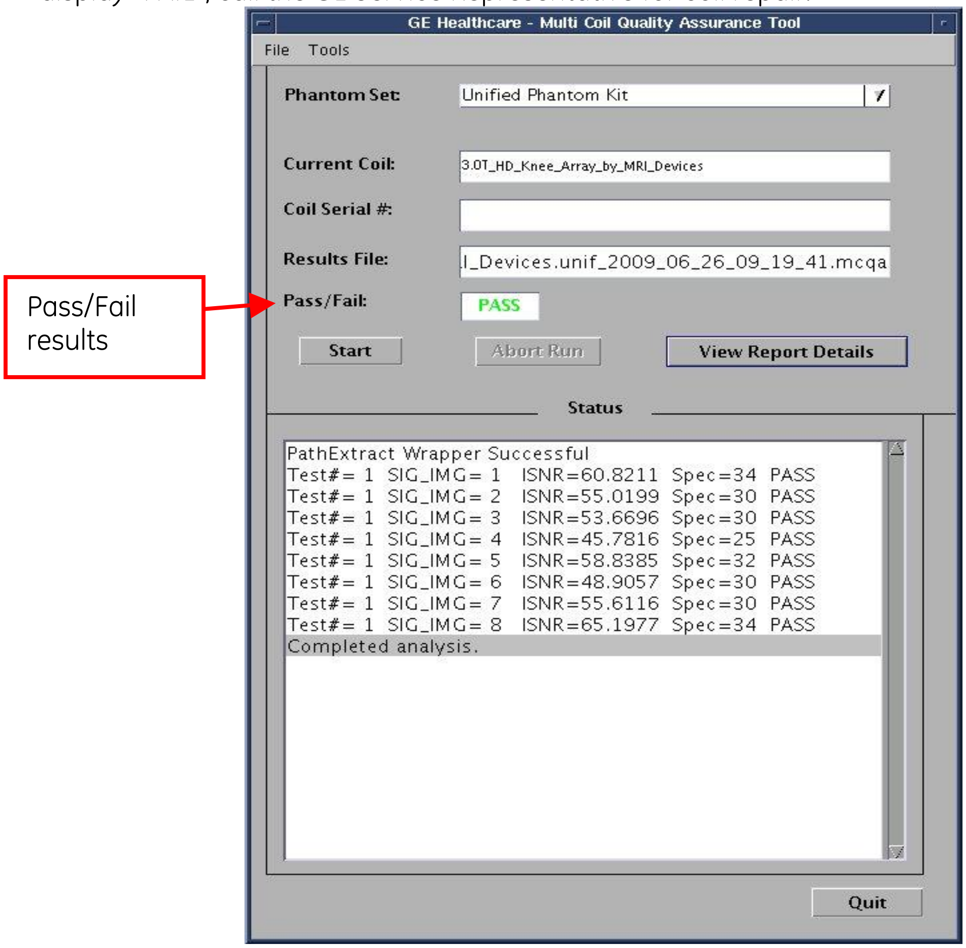


Figure 40

10. The MCQA Tool GUI can display “Fail” for several reasons. Possible reasons are, but are not limited to:
    - Bad Coil Element
    - Incorrect phantom used for the test
    - Incorrect positioning/placement of the phantom

11. Click on **[Quit]** button to exit MCQA Tool.

4 3T HNS Coil Troubleshooting Tips

4.1 Personnel Requirements

The following procedure requires one person for about 30 minutes.

4.2 Overview

The following tips can be used to troubleshoot common problems with the 3T HNS coil.

4.3 Preliminary Requirements

4.3.1 Tools and Test Equipment

- Some tests may require Phantoms for IQ Testing
- Digital Multi Meter

4.3.2 Replacement Parts

See Section 5:

4.3.3 Required Conditions

The following coil configurations must be installed to run the MCQA test for the coil:

Components Connected	Configurations
Posterior + Anterior	HNS HEAD
Posterior + NCU + Horseshoe Adapter + TL	HNS BrachPlex
TL only	HNS CTLBOT

Refer to the operator manual for all coil configurations.

4.4 Troubleshooting

4.4.1 Receiving No Signal

Problem:
You are unable to pre-scan or are scanning, yet receiving no signal, or receiving partial signal from coil elements.

Possible Solution:

- Verify that the port(s) that the coil is plugged into has a green light illuminated (Port A or Port B) or a green lock (Port P1, P2, P3 or P4) indicated. This indicates the coil is properly plugged into the system and the coil I.D. on the coil is properly installed.

  NOTE:
  - For Forward Production systems, use Port A (Hypertronics, 8 channels) for TL Unit (Thoracic Lumbar), and Port B (Hypertronics, 16 Channel) for Head Neck Unit. If the TL is used by itself, then it can also be connected to port B.

- o For upgrade systems: use Port A (Hypertronics, 8 channels) for TL (Thoracic Lumbar) Unit, Port B (Hypertronics, 8 Channel) and 8 channel legacy 3T connector for Head Neck Unit.
- o For Discovery systems, use Port P1 or Port P2 (P-connector, 16 channel) for HNU and use Port A (Hypertronics, 8 channel) for TL Unit.

- Verify that the system has correctly detected the coil by checking in the "Currently Connected" in the GUI to select coils in Rx as shown in Figure 41.

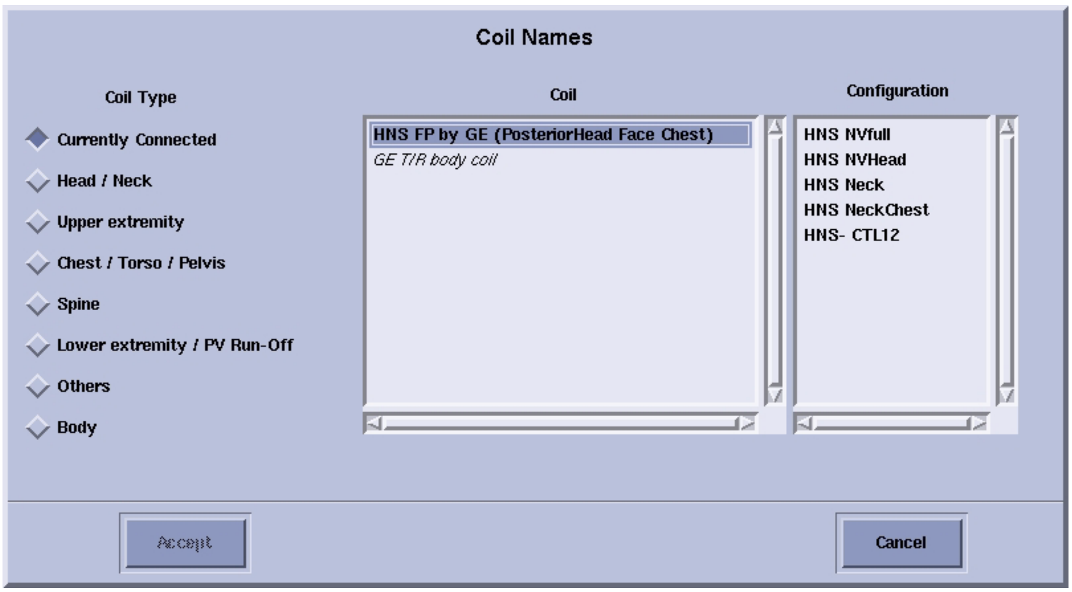


Figure 41: "Currently Connected" Coils Window In The Scan Screen

- Verify that the landmark is correct and that the cradle (patient bed) has not unlatched.
- Verify that the scan locations and any FOV offsets are correct.
- Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the damaged cable.
- Verify that the coil is positioned with the cable exiting towards the bore.

If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still receive no signal the problem probably lies with the MR system. If the scan completes successfully, there is probably a problem with the HNS coil. Contact GE for further assistance.
You may also try to scan with a similar type (receive only) coil. If you are unable to scan with this coil, there may be a system problem related to this particular coil type.

4.4.2 Image Quality

Problem:
Poor IQ, shaded images, or if MCQA fails

Possible Solution:

- Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the damaged cable.
- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.
- Verify that your center frequency is within the frequency adjustment range for your system.
- Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.

If you have not done so already, perform a system Quality Assurance MCQA test. If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using both another coil of the same type (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the image quality is visibly improved, there may be a problem with the Head Neck Spine coil. Contact GE for further assistance. If the image quality still suffers, there may be a system problem related to imaging with this type of coil.

4.4.3 Artifacts

Problem:
There is a black line, strip lines, or signal void on the image.

Possible Solution:

- Verify that there is no metal present in the area being scanned in, or on the patient.
- Verify that no hair is trapped in the connectors between the anterior and posterior sections of the head unit. This may appear as intermittent oscillations.
- Verify the cable connector of the NCU is fully engaged into the receptacle on the posterior section of the head unit. If the connector is not fully engaged, there may be a coil not present error or there may be intermittent oscillations.
- If you have oscillation of the preamp you will see striped lines.

5 3T HNS System Cable Replacement

5.1 Personnel Requirements

Required Persons: 1
Procedure Timing (mins): 60

Follow this process to change the system cable for coils that have a cable available as a spare part. This process will be the same whether replacing the coil or the cable

5.2 Tools and Test Equipment

Item: Standard tool kit – Flathead screwdriver, and pliers
Qty: 1

5.3 Replacement Parts

GEHC Part #	Description
5341961	HNU MR750 System Cable (w/ Coil ID Board)
5342130	HNU HDxT FP System Cable (16Ch) (w/ Coil ID Board)
5342132	HNU HDxT Upgrade System Cable (8CH) (w/ Coil ID Board)
5342134	HNU HDxT Upgrade System Legacy Cable (Bendix)
5342128	TL System Cable

5.4 Safety

No additional Safety Information is required

5.5 Procedure

5.5.1 ESD Notification and Coil Covers Removal

NOTICE

Please refer to the cable FRU list above in Replacement Parts.

- If the cable FRU is with the coil id board, then the coil id board **has to be removed** as well (HNU Only).
- If the cable FRU is without the coil id board, then **do not** remove the coil id board (TL Only).

NOTICE

Observe anti-static precautions: Use wrist strap and ESD workstation (ESD mat connected to ground and place to connect ESD wrist strap)

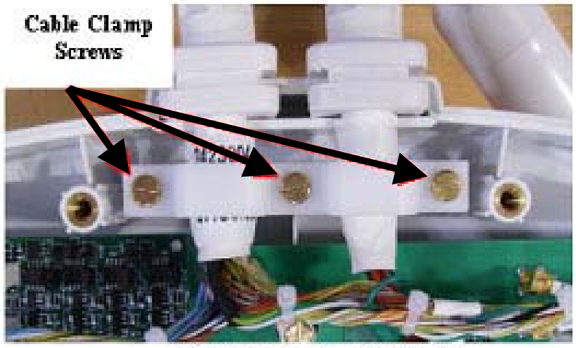
5.5.2 Put on a grounded ESD wrist strap.

Figure 42: ESD Wrist Strap On

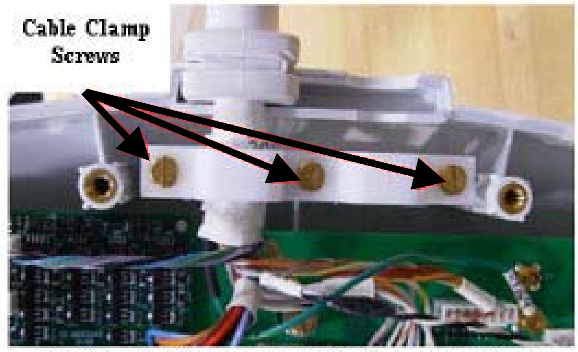
5.5.3 Unscrew the bottom cover and remove it from coil. Retain the original screws used.

5.5.4 Remove Cable Clamp

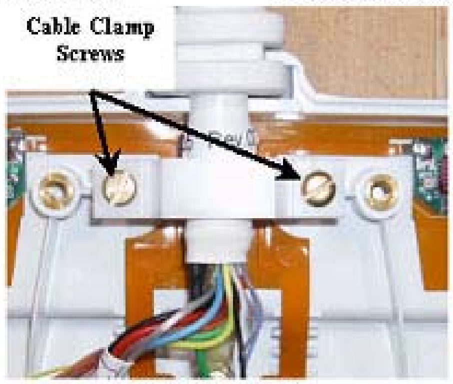
Unscrew the cable clamp from the coil as shown in Figure 43. Retain the original
Screws.



HNU Upgrade Illustration



HNU Forward Production & Discovery Illustration



TLU Illustration

Figure 43: Removing the Cable Clamp

5.5.5 *Unplug output cable from Interface Board*

Unplug the ganged RF connectors (black housing) and DC Headers of the output cable from the Interface board and Coil ID board. Use pliers to remove the PCX connector (grey arrow). **Note:** MR750 cable shown, for Upgrade systems there will be two cables that have additional connections (white arrows). See Figure 44.

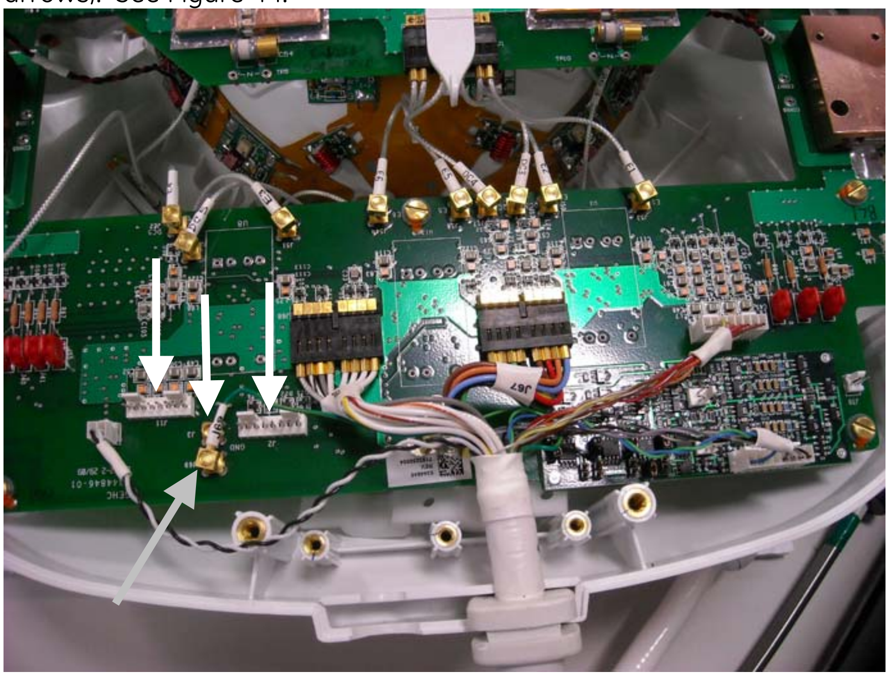


Figure 44: Unplugging Cable from HNU

5.5.6 *To remove cable clamp on TL, lift up on this side of clamp (grey arrow 1) and slide back (black arrow). See Figure 45.*

5.5.7 *Use pliers to remove the PCX connector (grey arrow2). See Figure 45.*

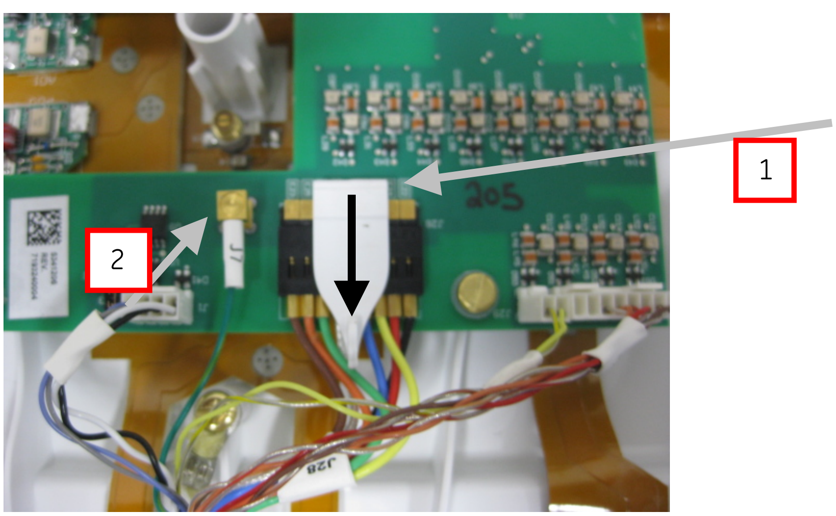


Figure 45: Unplugging Cable from TL

5.5.8 Unscrew the Kevlar Termination Screw

Remove the screw securing the Kevlar I-loop to the former. See Figure 46 (white arrow), or for the TL see Figure 47 (black arrow).

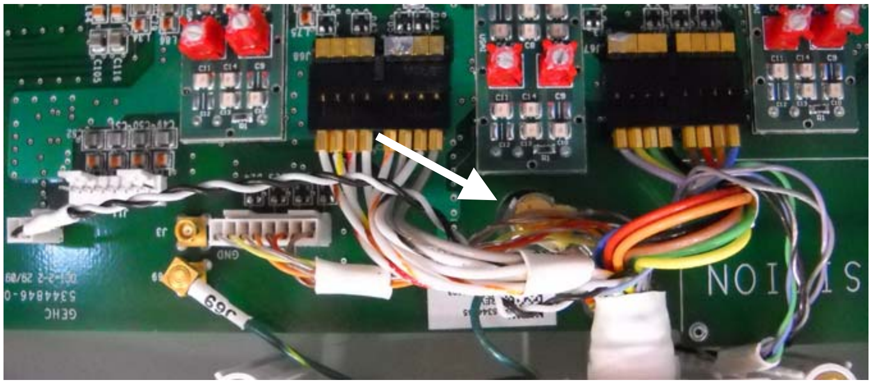


Figure 46: Unscrewing Kevlar Strap from HNU

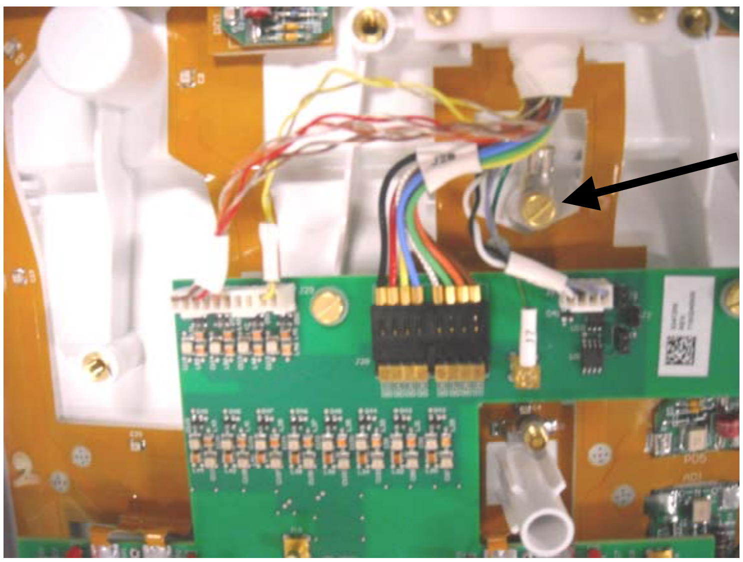


Figure 47: Unscrewing Kevlar Strap from TL

5.5.9 Remove the faulty cable from housing

5.5.10 Coil ID Board removal (HNU Only)

The coil ID board (outlined in white) is NOT soldered to the Interface board. It is mounted on. After removing both J55 and J56 (2-pin headers) (black arrows) from the coil ID board, the coil ID board can be removed by pulling the board up. See Figure 48 for a photograph of the ID board and Figure 49 for a photograph of after the ID board is removed.

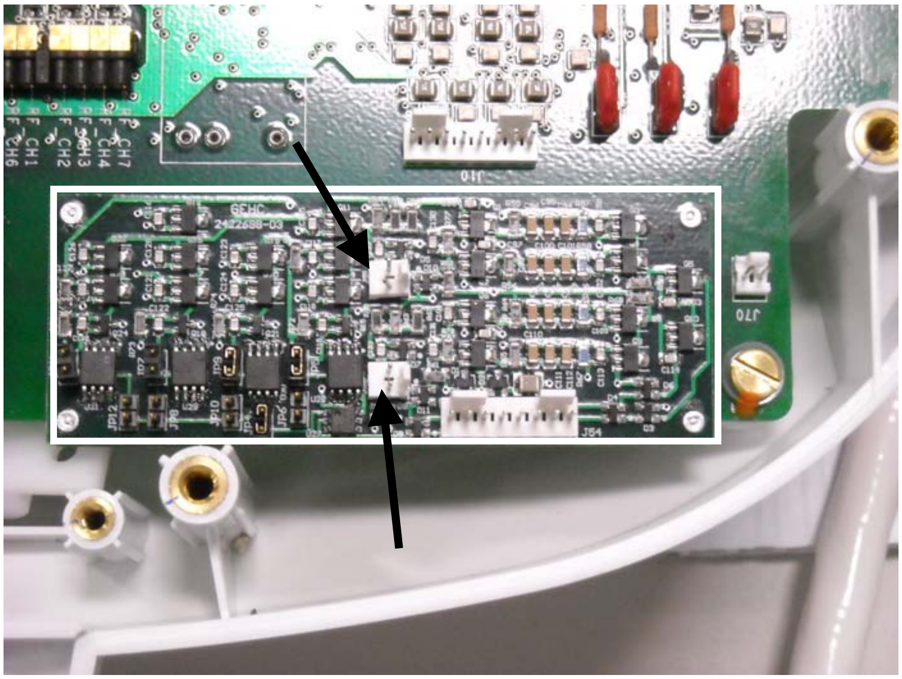


Figure 48: Coil ID Board Location

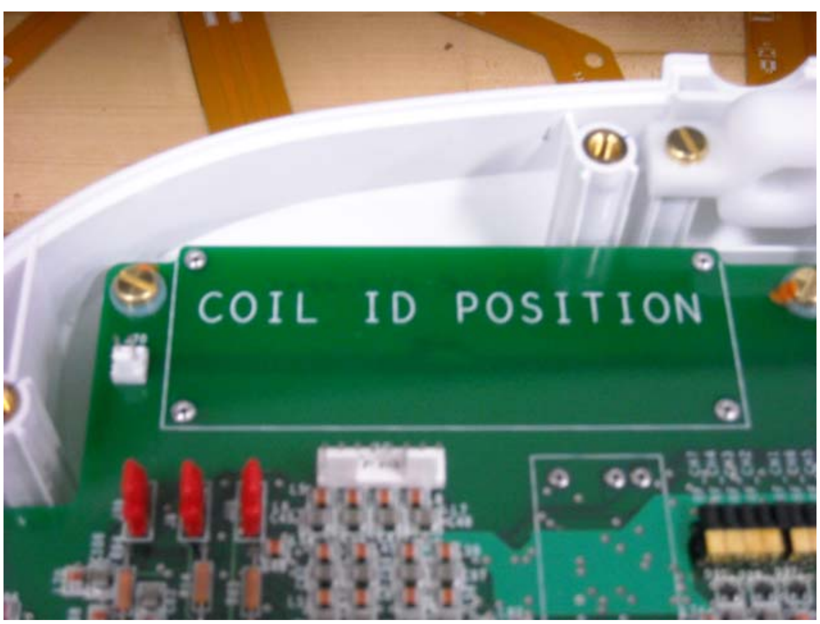


Figure 49: Coil ID Board Removed

5.5.11 Install new Coil ID Board (HNU Only)

Install Coil ID board into "Coil ID Position" on the interface board. Connect the 2-pin header from J70 to J55 (black arrow), and from J112 to J56 (dashed black arrow).

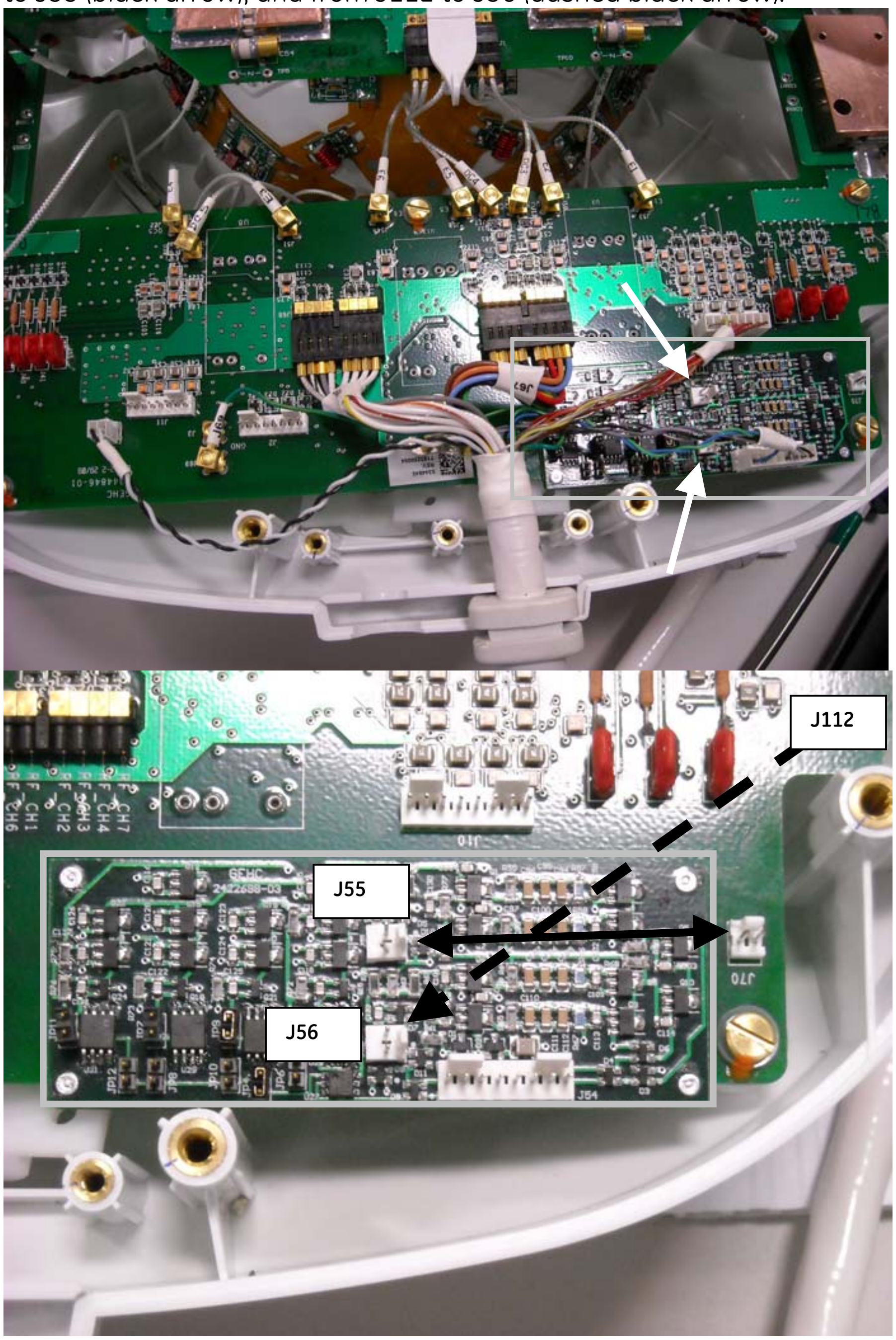


Figure 50: Coil ID Board Location

5.5.12 Install new Cable into Cable Clamp

Place cable into housing. Secure cable clamps around the cables being sure not to pinch cables. Tighten screws to 10In-lbs.

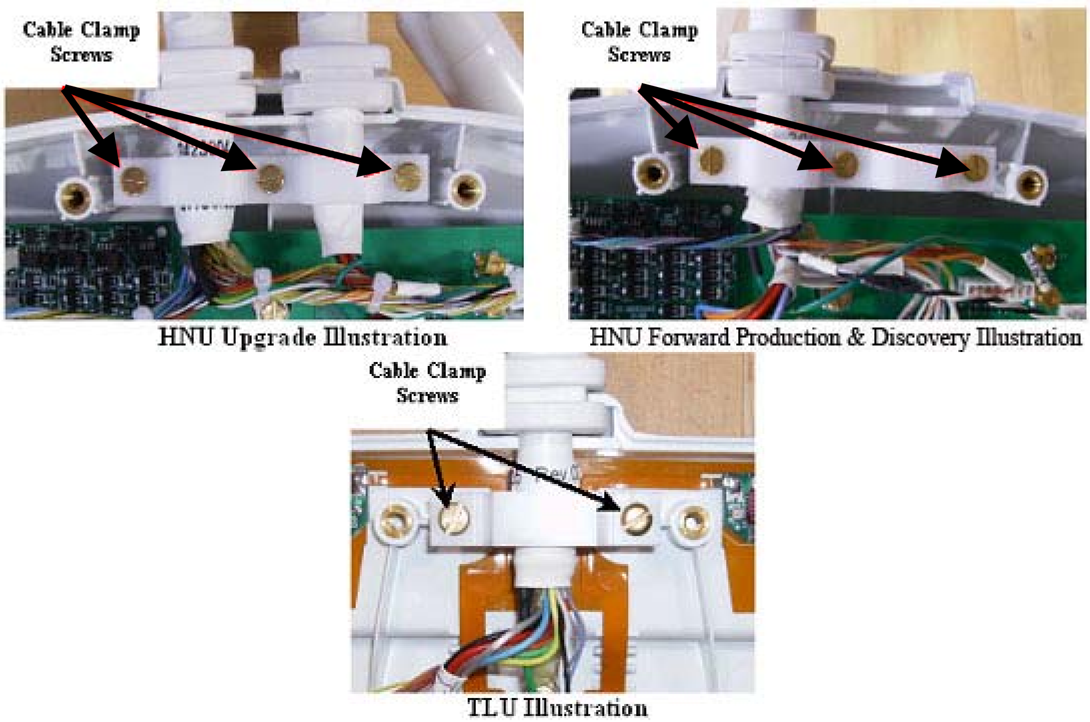


Figure 51: Cable Clamp Install

5.5.13 Secure Kevlar Loop

Secure the Kevlar I-loop to the HNU (Figure 52) (white arrow) or the TL (Figure 53) (black arrow) as shown. Use a 10in-lbs torque driver to tighten screws.

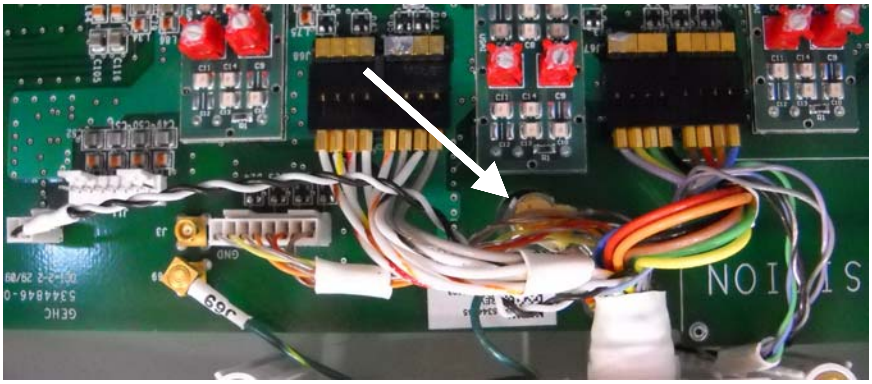


Figure 52: Re-install Kevlar Strap from HNU

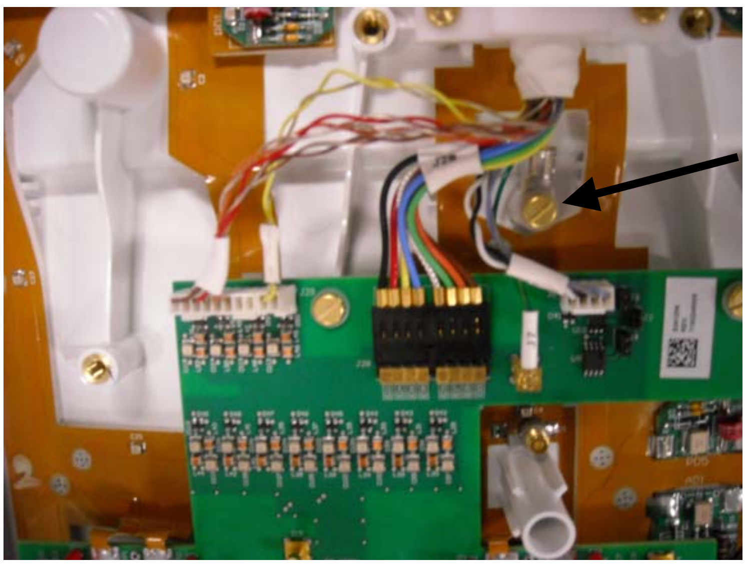


Figure 53: Re-install Kevlar Strap from TL

5.5.14 Connect Cables

Connect the cables to the interface board by matching the cable label to the silkscreen on Interface PCB.

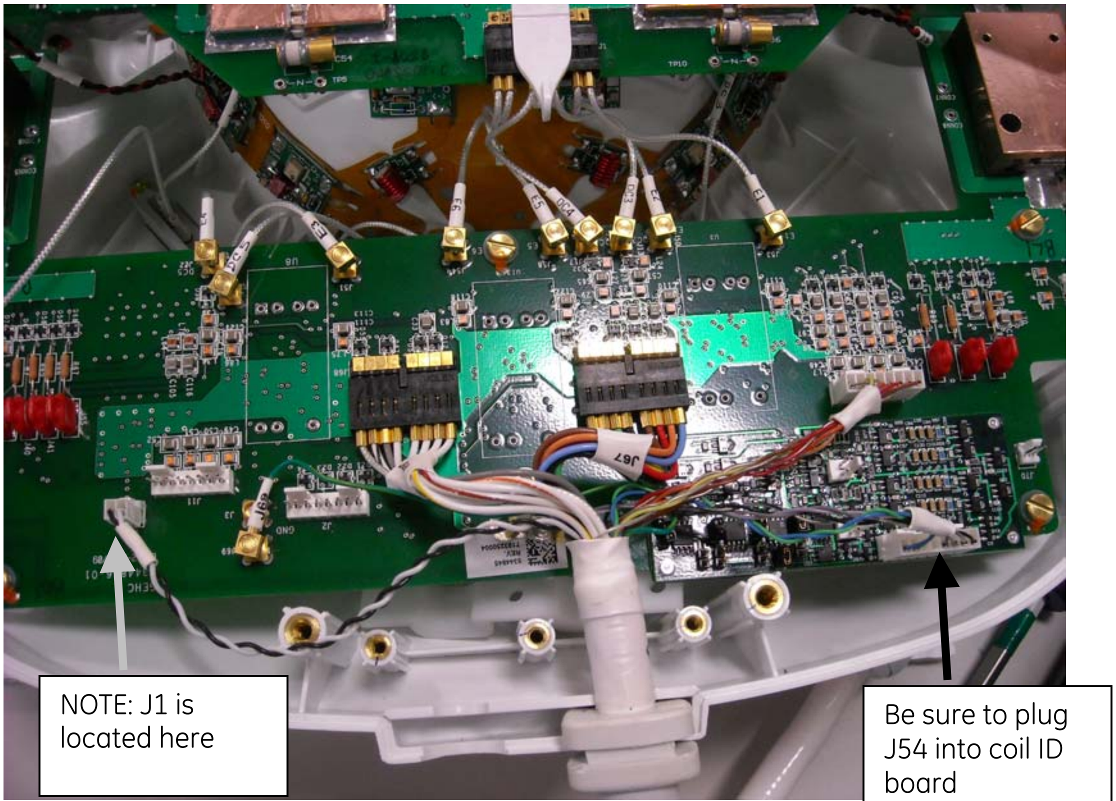


Figure 54: Connect system cables to HNU

5.5.15 Attach cables to TL feedboard (Silkscreen on board matches cable). Re-install the cable clamp (Use black arrows for guidance) on the ganged connector (black connector). See Figure 55.

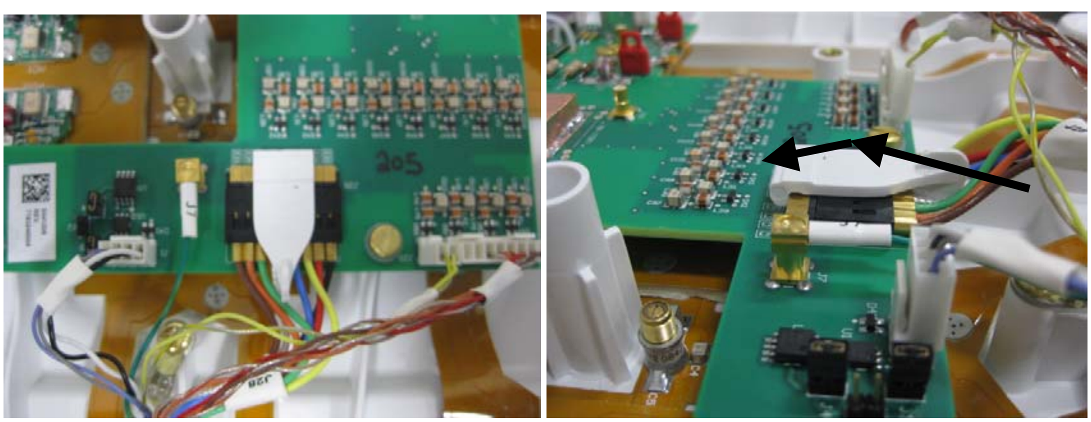


Figure 55: Attach cables to TL feedboard and re-install the cable

5.5.16 For Head Neck Unit Field Replacement Only

If the Head Neck Unit has been replaced and is being installed into a Forward Production HDxT system, remove red suitcase jumpers located at J5, J40, J41, J55, J28, J8, and J39. Keep the 7 jumpers in place for Upgrade and MR750 systems. The jumpers are shown in the grey boxes in Figure 56.

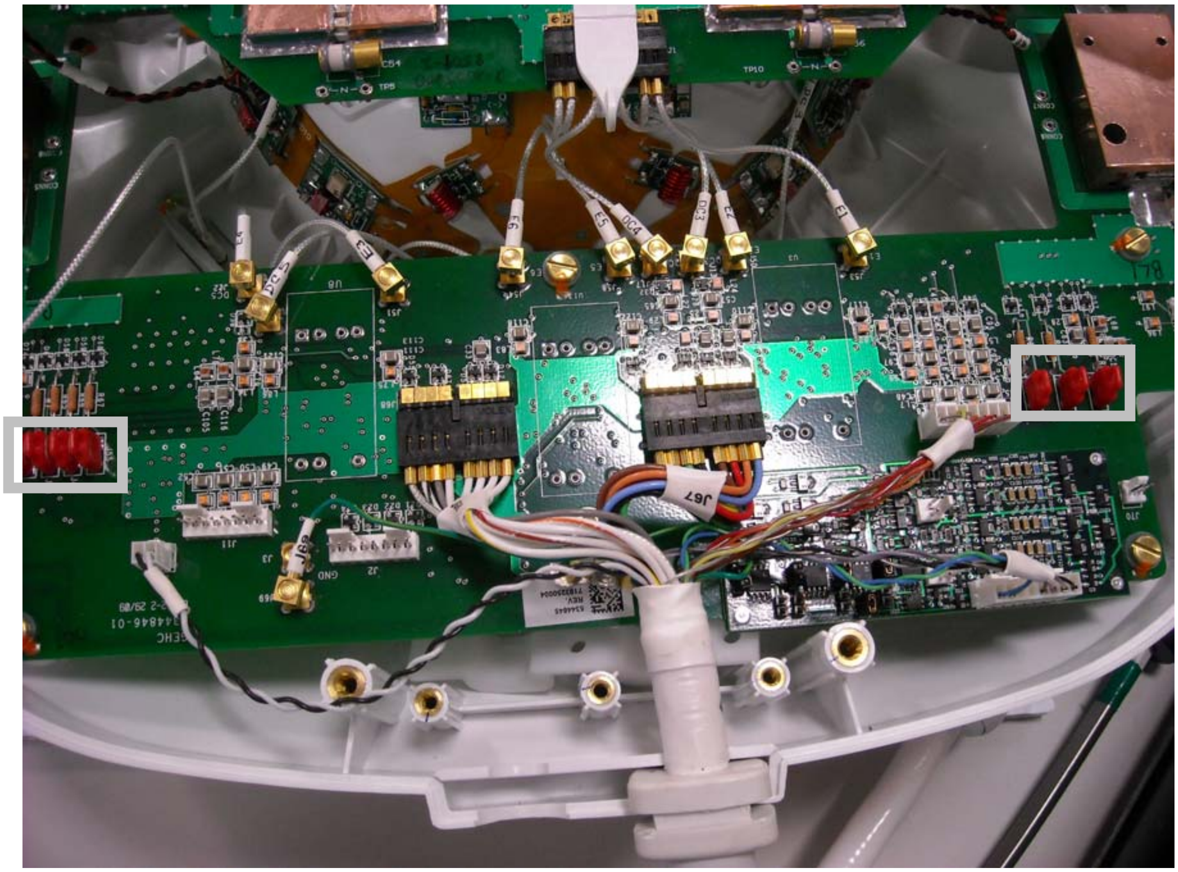


Figure 56: Jumper Illustration

5.5.17 Replace the cover of the coil. Use original screws to fasten cover.

5.6 Finalization

Perform a MCQA scan with the coil to confirm that the repaired coil operates properly.

6 3T HNS Coil Configuration for Reverse Ramped System

6.1 Personnel Requirements

Required Persons: 1
Procedure Timing (mins): 45

Follow this process to make the coil compatible with reverse ramped system.

6.2 Tools and Test Equipment

Item: Standard tool kit – Pliers
Qty: 1

6.3 Replacement Parts

No Parts need to be replaced

6.4 Safety

No additional Safety Information is required

6.5 Procedure

6.5.1 ESD Notification

Observe anti-static precautions: Use wrist strap and ESD workstation (ESD mat connected to ground and place to connect ESD wrist strap)

6.5.2 Put on a grounded ESD wrist strap.

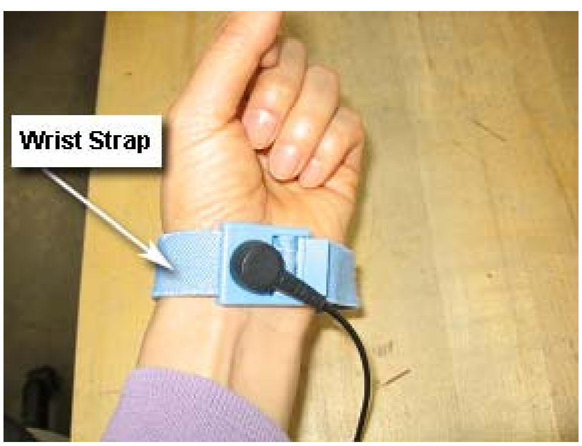


Figure 57: ESD Wrist Strap On

6.5.3 *Connection swap*

Unscrew the posterior cover of the HNU and remove from coil. Retain the original screws used.

The coil by default is configured for Forward Ramped Systems. For G3 magnet reverse ramped systems, the connections on the MUX sub-assembly should be changed as shown in Figure 58, Figure 59, and Figure 60.

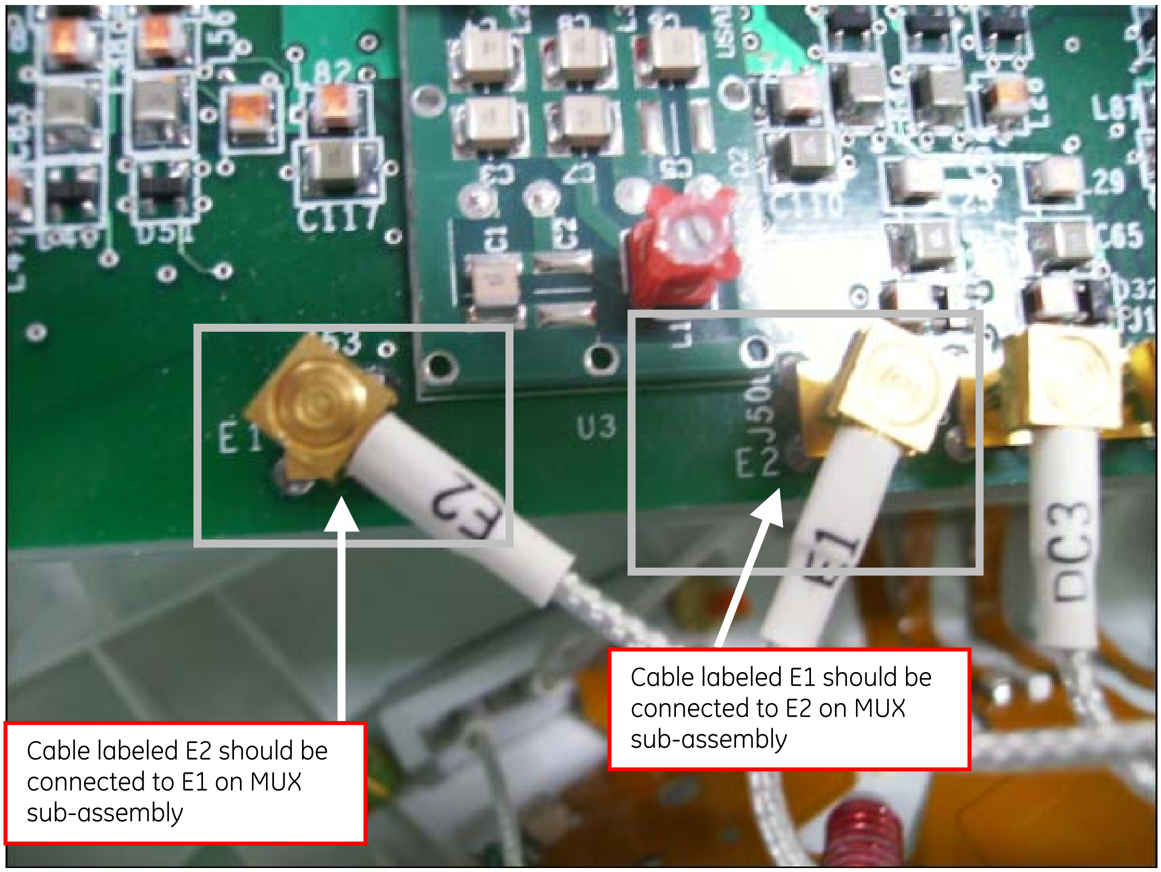


Figure 58: The cable labeled E2 should be connected to E1 on MUX sub-assembly, and the cable labeled E1 should be connected to E2 on MUX sub-assembly

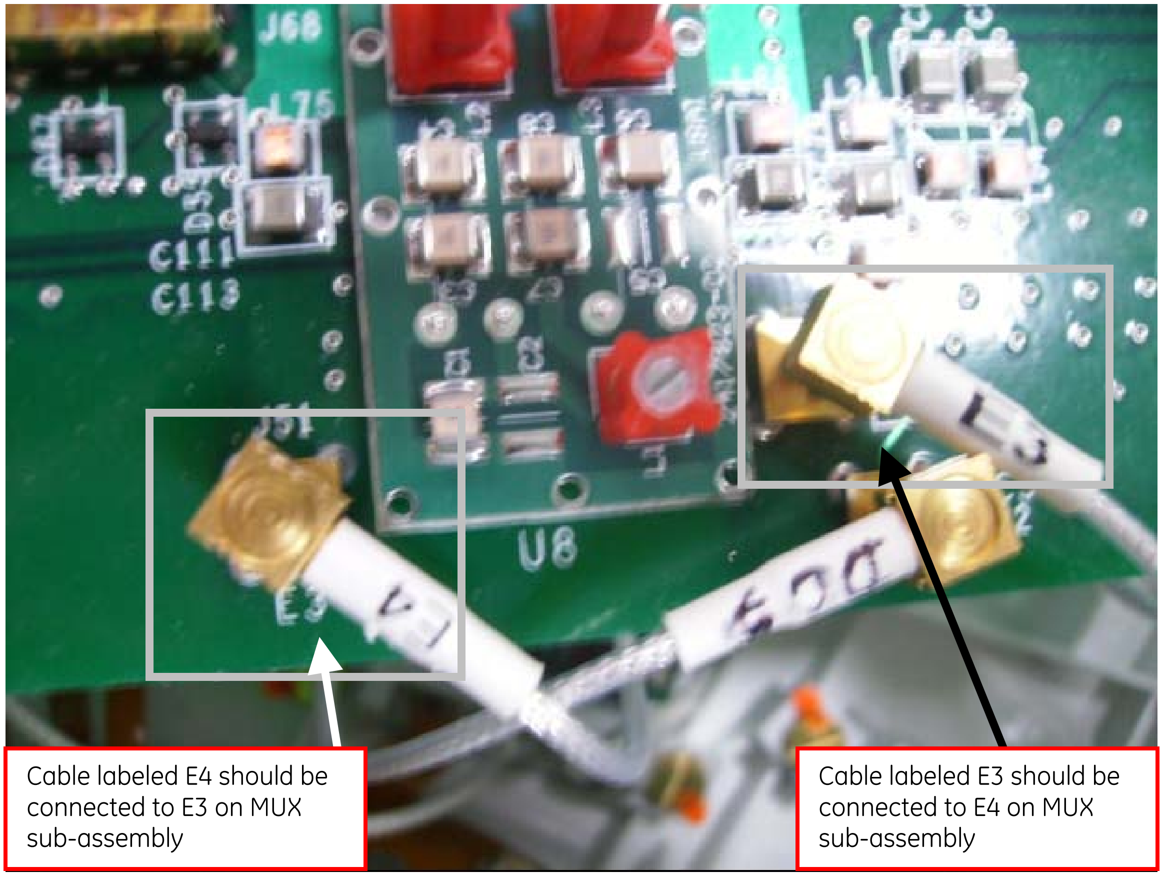


Figure 59: The cable labeled E4 should be connected to E3 on MUX sub-assembly, and the cable labeled E3 should be connected to E4 on MUX sub-assembly

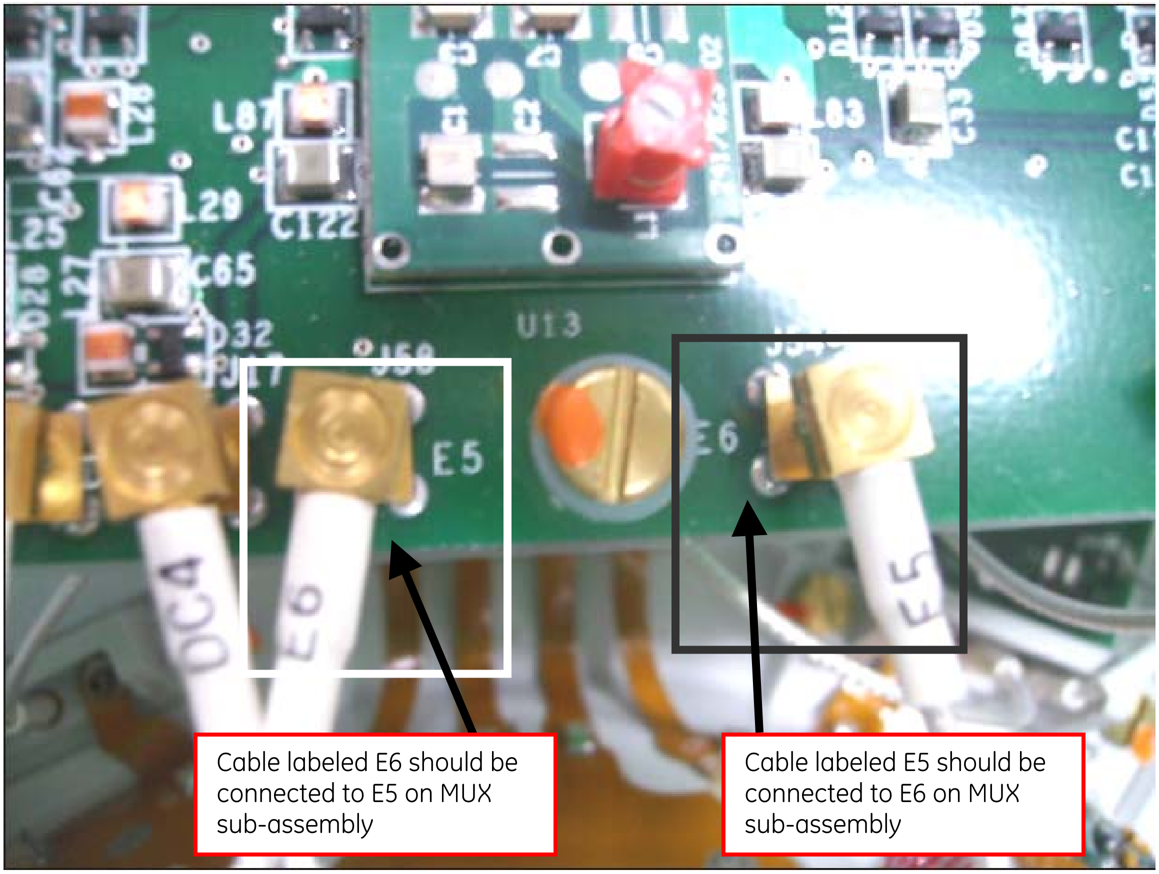


Figure 60: The cable labeled E6 should be connected to E5 on MUX sub-assembly, and the cable labeled E5 should be connected to E6 on MUX sub-assembly

6.5.4 Replace the posterior cover of the HNU. Use original screws to fasten cover.

6.6 Finalization

Perform a MCQA scan to ensure that the coil works after the connection swap.

7 3T HNS Coil Field Replaceable Units

Table 2: 3T HNS FRU's

NOTE: All pieces should be visually inspected for damage (i.e. bent pins, holes, cracks) prior to use.

Part No.	Description	Picture
5338925	3T Head Neck Unit - Posterior, Anterior, Horseshoe adapter	
5338938	3T HNS Neck Chest Unit	
5336485	3T HNS TL Unit	
2381682-10	3T HNS TL Unit Phantom (Includes one large brick SiOil Phantom)	

5366263	3T HNS Head Neck Unit Phantom Kit (Includes one cylinder and one brick shaped SiOil phantom)	 
5366262	3T HNS Head Neck Unit Phantom Positioner	
2419521	3T HNS Mirror	
5341961	HNU MR750 System Cable (Includes coil ID Board – not shown)	

5342130	HNU HDxT FP System Cable (16Ch) (Includes coil ID Board – not shown) **NOTE:** 16CH Cable has 2 sets of ganged connectors (see arrows at photograph to the right)	
5342132	HNU HDxT Upgrade System Cable (8CH) (Includes coil ID Board – not shown) **NOTE:** 8CH Cable has 1 ganged connector (see arrow at photograph to the right)	
5342134	HNU HDxT Upgrade System Legacy Cable (G3)	
5342128	TL System Cable	
5365916	Hardware Kit	Not Applicable

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Imagination at work